

Cancer Classification Based on Microarray Gene Expression Data Using Deep Learning

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Abstract—The classification of cancer is a major research topic in Medicine. Cancer microarray data normally contains a small number of samples, which have a large number of gene expression levels as features, however, makes the classification quite challenging. Using a deep learning algorithm based on multilayer perceptron, we show that classification performance at least as good as published results can be obtained for cancer classification.

Keywords—cancer; classification; microarray gene expression; machine learning; deep learning

I. INTRODUCTION

Biomedical information processing mechanism suggests the need of deep architectures for extracting patterns that allows characterize with highest accuracy the system under study. It is natural to believe that the state of the art can be advanced in processing of biomedical systems if efficient and effective deep learning algorithms are developed. The concept of deep learning originated from artificial neural network research. Multilayer perceptron with many hidden layers is a good example of the models with deep architectures [1]. Deep learning techniques have been applied to a wide variety of problems in recent years [2, 3]. In many of these applications, algorithms based on deep learning have surpassed the previous state-of-art performance. At the heart of all deep learning algorithms is the domain independent idea of using hierarchical layers of learned abstraction to efficiently accomplish high-level task. Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction. Deep learning discovers intricate structure in large data sets by using the backpropagation algorithm to indicate how a machine should change its internal parameters that are used to compute the representation in each layer from the representation in the previous layer [1].

Cancer is a heterogeneous complex of diseases, with distinct biological features that lead to differences in response patterns to various treatment modalities and clinical outcomes. Tumors with similar clinical and pathological presentations may have different behaviors. Gene expression analysis have focused on defining more detailed biological characteristics to improve patient risk stratification and to ensure the highest chance of benefit and the least toxicity from a specific treatment

modality. Gen signatures have been developed as predictors and now hold promise for the development of more accurate test for the classification and the prediction of recurrence of cancer. Several machine learning algorithms have already been applied to classifying tumors using microarray data [4, 5, 6, 7]. Accurate cancer diagnosis is vital for the successful application of specific therapies. Although cancer classification has improved over the last decade, there is still a need for a fully automated and less subjective method for cancer diagnosis. Given the nature of cancer microarray data, which usually consists of a few hundred samples with thousands of genes as features, the analysis has to be carried out carefully. There are many approaches and procedures designed to detect group of genes that are strongly associated with a particular cancer type. Deep learning, by contrast, apply machine learning algorithms to feature subsets and use cross-validation to evaluate the score of feature subsets. Deep learning use classifiers to estimate the usefulness of feature subsets, which should provide a better classification accuracy for the corresponding classifiers, since the features are selected according to their contribution to the classification accuracy of the classifiers.

The rest of this paper is organized as follows. We begin with a brief introduction to deep learning, following by a description of the cancer data base under study. Thereafter, we present the experimental results, and finally, the last section discusses the results and conclude this paper.

II. MATERIALS AND METHODS

A. Deep Learning

The concept of deep learning originated from artificial neural network research. Multilayer perceptron with many hidden layers is a good example of the models with deep architectures. Unlike the neural networks of the past, modern deep learning has cracked the code for training stability and generalization and scale on big data. It is often the algorithm of choice for highest predictive accuracy, as deep learning algorithms performs quite well in a number of diverse problems. There are several theoretical frameworks for deep learning, and here we summarize the feedforward architecture used by H20 [8]. Multilayer perceptron (MLP) are feed-forward neural networks with architecture composed of the input layer, the hidden layer and the output layer. Each layer is

formed from small units known as neurons. Neurons in the input layer receive the input signals X and distribute them forward to the rest of the network. In the next layers, each neuron receives a signal, which is a weighted sum of the outputs of the nodes in the previous layer. Inside each neuron, an activation function is used to control the input. Such a network determines a non-linear mapping from an input vector to the output vector, parameterized by a set of network weights, which are referred to as the vector of weights W . The first step in approximating the weight parameters of the model is finding the appropriate architecture of the MLP, where the architecture is characterized by the number of hidden units, the type of activation function, as well as the number of input and output variables. The second step estimates the weight parameters using the training set. Training estimates the weight vector W to ensure that the output is as close to the target vector as possible. The structure of a MLP network is shown in “Fig. 1”. This basic framework of MLP neural networks can be used to accomplish deep learning task. Deep learning architectures are models of hierarchical feature extraction, typically involving multiple levels of nonlinearity.

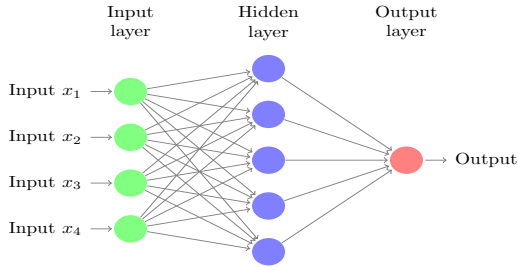


Fig. 1. Structure of an architecture multilayer perceptron

B. Dataset

One of the most famous microarray databases is the 11 Tumors database. This database is available freely online (<http://www.gems-system.org>), and is the most commonly used database by researches in this field. This data consists of 174 samples with 12,533 genes, and the samples belong to a variety of classes: 0 belongs to Ovary class, 1 belongs to Bladder/Ureter class, 2 belongs to Breast class, 3 belongs to Colorectal class, 4 belongs to Gastroesophagus class, 5 belongs to Kidney class, 6 belongs to Liver class, 7 belongs to Prostate class, 8 belongs to Pancreas class, 9 belongs to Lung Adeno class, 10 belongs to Lung Squamous class.

C. Supervised Classification of Cancer Using Machine Learning

In supervised learning we assume each element of study is represented as an n -component vector-valued random variable $(X_1, X_2, \dots, X_n)$, where each X_i represents an attribute or feature; the space of all possible feature vectors is called the input space X . We also consider a set $\{w_1, w_2, \dots, w_k\}$ corresponding to the possible classes; this forms the output space W . A classifier or learning algorithm typically receives, as input, a set of training examples from a source

domain $T = (x_i, w_i)$, where; $x = (x_1, x_2, \dots, x_n)$ is a vector in the input space, and w is a value in the (discrete) output space. We assume the training or source sample T consists of independently and identically distributed (i.i.d.) examples obtained according to a fixed but unknown joint probability distribution, $P(x, w)$, in the input-output space. The outcome of the classifier is a hypothesis or function $f(x)$ mapping the input space to the output space, $f: X \rightarrow W$. We commonly choose the function that minimizes the expected value of a loss function (e.g., number of misclassifications).

D. Tools

Library H20 [8] was used in order to perform the classification through deep learning.

III. RESULTS

In our approach, we train a MLP on a set of randomly selected samples, approximately 60 %, extracted from the dataset, then approximately 20 % are used as the validation set, and approximately 20 % are used as the testing set. The purpose of this training is to learn the multilayer architectures by simple stochastic gradient descent [1]. Table 1 shows the size of the architecture multilayer perceptron and parameters used on the experiments to evaluate the classification. Using these parameters allowed achieve higher classification accuracy. ReLU is the non-linear activation function, 11 and 12 are parameters of regularization to avoid overfitting, epochs correspond to the numbers of passes over the training data set, and nfolds correspond to cross-validation.

TABLE I. PARAMETERS OF THE ARCHITECTURE MULTILAYER PERCEPTRON

Variables	Parameters
input	174
hidden	(50,50,50)
output	11
Activation function	ReLU
Loss function	Mean Squared Error
Epochs	1000000
11	0.00001
12	0.00001
nfolds	10

TABLE II. PRECISION REPORTED ON TRAIN DATA

Deep learning
** Reported on train data. **
MSE: 1.38526e-11
R ² : 0.9999

TABLE III. PRECISION REPORTED ON VALIDATION DATA

TABLE IV.

Deep learning
** Reported on
validation data. **
MSE: 0.15
R ² : 0.9864

We can see that the results in the training set and validation set are consistent with the architecture and variables used for classification. Results reported in Table II are encouraging with the deep learning approach for classification of Tumors under analysis.

IV. CONCLUSIONS

In this work, we have presented an application of supervised machine learning for classification of several types of cancer via deep learning. First, we showed that the network could learn surprisingly well. Second, we showed that the network can classify the different samples with high efficiency, and finally, a high precision is achieved in the task of predicting the different classes.

Our experiment indicate that a deep learning approach in combination with gene microarray features, has the potential to capture gene expression patterns, which may boost the clasification performance. We conclude that machine learning techniques, like deep learning, in combination with new molecular techniques, hold promise for improving diagnosis, better assestment of recurrence risk, careful selection of therapy and identification of targeys involved in carconogenesis and function of tumors cell, leading to the discovery of selective drugs.

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